Statistics

The big idea

When data is grouped into class intervals (0–10, 10–20, …), we can't see each value, so we estimate the **mean**, **median** and **mode** using formulas. Each tells a different story: the average, the middle and the most common.

Mean of grouped data

For each class, take its **class mark** x_i = (lower limit + upper limit)/2.

THREE WAYS TO FIND THE MEAN

Direct: $\bar{x} = \Sigma f_i x_i / \Sigma f_i$

Assumed mean: $\bar{x} = a + \Sigma f_i d_i / \Sigma f_i$, where $d_i = x_i - a$

Step deviation: $\bar{x} = a + (\Sigma f_i u_i / \Sigma f_i) \times h$, where $u_i = (x_i - a)/h$

All three give the same answer; the step-deviation method keeps the numbers small.

Mode of grouped data

The **modal class** is the class with the **highest frequency**.

MODE

$$\text{Mode} = l + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$$

l = lower limit of the modal class, h = class size, f_1 = frequency of the modal class, f_0 = of the class before it, f_2 = of the class after it

Median of grouped data

Make a **cumulative frequency** column. Find n/2; the class whose cumulative frequency first reaches n/2 is the **median class**.

MEDIAN

Median = l + [(n/2 − cf) / f] × h

cf = cumulative frequency of the class before the median class, f = frequency of the median class

FIND THE MEDIAN: 0–10: 5, 10–20: 8, 20–30: 20, 30–40: 15, 40–50: 7, 50–60: 5

n = 60, n/2 = 30. Cumulative: 5, 13, 33 … so the median class is 20–30.
l = 20, cf = 13, f = 20, h = 10. Median = 20 + (30 − 13)/20 × 10 = 20 + 8.5 = **28.5**.

EMPIRICAL RELATIONSHIP

3 Median = Mode + 2 Mean

USING CF OF THE MEDIAN CLASS ITSELF

In the median formula, cf is the cumulative frequency of the class **just before** the median class.

Quick recap

- Mean uses class marks; choose direct, assumed-mean or step-deviation.
- Mode: modal class = highest frequency, then the formula.
- Median: cumulative frequency, median class at n/2, then the formula.

Practice questions

1. The class mark of the class 10–25 is
   (a) 15 (b) 17.5 (c) 12.5 (d) 35
2. The modal class is the class with the
   (a) largest class mark (b) highest frequency (c) highest cumulative frequency (d) middle position
3. If mode = 12 and mean = 15, then the median (empirical relation) is
   (a) 13 (b) 14 (c) 16 (d) 18
4. To find the median of grouped data we first need
   (a) class marks (b) cumulative frequencies (c) deviations (d) the modal class
5. In the step-deviation method, u_i equals
   (a) $x_i - a$ (b) $(x_i - a)/h$ (c) $f_i x_i$ (d) x_i/h

Answers: 1. b 2. b 3. b 4. b 5. b